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ORIGINAL PAPER

In vitro Cytotoxicity and Antibacterial Activity of Optimized Silver Nanoparticles Against Wound Infectious Bacteria and Their Morphological Studies

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Abstract

The present study reports the optimized silver nanoparticles (AgNPs). They were tested against, three wound infecting gram-positive and gram-negative pathogenic bacteria viz., *B. subtilis, S. aureus, M. luteus, E. coli, K. pneumoniae* and *P. aeruginosa* at 10 µg concentration. In the antibacterial activity of optimized AgNPs, *P. aeruginosa* was found remarkably sensitive to the AgNPs with 20.3 ± 1.86 mm of inhibition zone. The standard antibiotics of streptomycin, gentamycin, ampicillin and erythromycin at 10 µg, when tested against the bacteria, which revealed that gentamycin showed high antibacterial activity against all the six wound infecting pathogenic bacteria. The MIC and MBC concentrations evaluated against the bacteria showed were 4.0 ± 1.00 µg/mL and 6.3 ± 0.47 µg/mL for *S. aureus*, respectively. The normal morphology of tested bacteria was changed when treated with optimized AgNPs as evidenced under TEM analysis. The optimized AgNPs impregnated on cotton fabrics showed high antibacterial activity against tested bacteria. Thereafter there was a gradual decrement in antibacterial activity against the bacteria at 2nd, 3rd and 4th wash. The cytotoxicity effect of optimized AgNPs treated normal cell (Vero) recorded a maximum IC_{50} value 20 µg/mL. Our research outcome opens up new improved antimicrobial composition in pharmaceutical and medical sectors.

Keywords Optimized AgNPs · Wound infectious bacteria · Antibacterial activity · AgNPs coated fabric · Cytotoxicity

Introduction

The significance of medicinal features of silver has been well known for past 2000 years. Silver nanoparticles (AgNPs) are exploited as a controlling agent in various unrestricted public areas and well-studied ion-based composites with extremely lethal to 16 major genera of bacterial pathogens [1]. Thus, the biogenic feature of AgNPs elicits an ideal choice for pharmaceutical and medical fields.

Due to the outbreak of pathogenic infections around the world, the needs for advanced nanomaterial have been proposed to the community to develop resistance against the diseases [2]. Among several noble nanoparticles like gold, copper, platinum, and zinc the silver nanoparticles have shown distinct properties of high anti-microbial potential, less toxicity, low volatility and high thermal

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stability since they are natural inorganic metal. Silver nanoparticles are broadly employed in diverse areas like textile fabrication, fabric cleaning, water treatment process, electronic or optical devices, sunscreen lotions, burn treatment and molecular diagnostic analysis by quantitative determination [3]. Silver nanoparticles have being used to making dental composites; in coating medical equipment; water filters with bactericidal covering materials. According to Mohanpuria et al. [4] "AgNPs as an antimicrobial agent in air spray sanitizers, pillows, respirators, socks, wet wipes, detergents, soaps, shampoos, toothpastes, washing machines and other products like bone cement, and wound dressings". Microorganisms are frequently infect human and causes diseases in the present living environments. Use of silver-based antimicrobial agent may be used against antibiotic resistance microbes [5].

In the existent study of optimized photosynthesized AgNPs were assessed their potential antibacterial activities against three wound infecting gram-positive (*B. subtilis, S. aureus, M. luteus*) and gram-negative bacteria (*E. coli, K. pneumoniae, P. aeruginosa*) were evaluated. The consistent minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC) data of the strains have been recorded. Sondi and Salopek-Sondi [6] recorded that AgNPs can cause cell lysis, inhibit cell transduction or may also lead to cell phosphorylation. The present attempt also aimed to check the stability of incorporated AgNPs in the cotton fabrics under four consecutive washing using a laundry detergent. The low-level coatings of optimized AgNPs in the fabrics create a durable antibacterial activity [7]. The cytotoxicity efficiency of varied concentrations of silver nitrate and optimized AgNPs were also tested on normal Vero cell line.

Recently, our team were synthesized AgNPs with *Cassia roxburghii* [8] and it was shown more stability, efficient activities against both plant and human fungal pathogens [9]. The optimized stable AgNPs-incorporated cotton fabrics were also shown the significant in vivo wound healing activities [10]. Henceforth, in this current study is intended to assess the ability of phytosynthesized AgNPs for the augmentation of enhanced antibacterial activity and their morphological changes.

Materials and Methods

Chemicals, Culture Media and Microorganisms

Muller Hinton agar (MHA) medium and antibiotic disk like streptomycin, gentamycin, ampicillin and erythromycin were bought from the HiMedia, Mumbai. Six different wound infection causing bacteria viz., *Bacillus subtilis, Staphylococcus aureus, Micrococcus luteus, Escherichia coli, Klebsiella pneumoniae* and *Pseudomonas aeruginosa* were obtained from culture collection centre at CAS in Botany, University of Madras, Chennai. Cell culture medium and antibiotic solutions were procured from the Sigma-Aldrich, USA.

Optimization and Analytical Characterization of AgNPs

Significantly stable AgNPs were synthesized with *C. roxburghii* leaves extract (1.0 mL) and it was optimized with different physicochemical conditions, (pH 7.0, 1.0 mM $AgNO_3$ and at 37 °C). The phytosynthesized AgNPs were examined with XPS, DLS and ZETA potential. Further, ZETA potential and DLS analysis, and found to be the average size of AgNPs were 35 nm with − 18.3 mV zeta potential. The sample preparation and instrument calibrations were accomplished according to our earlier report [9].

HRTEM, EDX and SAED Analysis of Optimized AgNPs

The optimized AgNPs were analysed their morphology through High-Resolution Transmission Electron Microscopy (HRTEM), Energy Dispersive X-Ray Analysis (EDX) and Selected Area Electron Diffraction (SAED). The AgNPs were analyzed with HRTEM, EDX and SAED the methods described by Balashanmugam and Thangavelu [8].

Atomic Force Microscopy (AFM)

The surface features and size of optimized AgNPs were analyzed by Atomic Force Microscope (AFM, XE-70 Park system, German) below standard atmospheric state. The working condition of AFM was fixed with F0 = 241 kHz—high-resonant-frequency, 41 N/m—silicon probes force constants, and 2 Hz scan speeds. The operation conditions were adjusted and fixed with AFM image software.

Antibacterial Activity of Optimized AgNPs Against Human Pathogenic Bacteria

The optimized AgNPs were evaluated for their antibacterial potential versus the wound infection causing three bacterial pathogens. The antibacterial activities of photosynthesized AgNPs were examined the agar well diffusion method described by Perez et al. [11]. Briefly, the bacterial pathogens were seeded uniformly with a cotton swab on Muller Hinton Agar (MHA) medium. The agar wells (6 mm) were made with sterile cork borer and then a, b, c, d labeled wells were loaded with 10 µg of AgNPs (40 µL), 10 μg of *C. roxburghii* leaves extract (40 μL), 10 μg of $AgNO_3$ (40 μL) and 40 μL of sterile distilled water (control) respectively. The plates were incubated for 24 h at 37 °C. In addition, standard discs *i.e.*, streptomycin (10 μg), gentamycin (10 μg), ampicillin (10 μg) and erythromycin (10 μg) were used as a positive control. The zone of inhibition (mm) was recorded after incubation periods.

MIC and MBC of Biosynthesized AgNPs Against Bacterial Pathogens

The dynamics of bacterial susceptibilities were recorded on the solid medium. The optimized phytosynthesized AgNPs were tested against all six bacterial pathogens. 5 mL of the optimally grown bacterial suspension was treated with AgNPs at different concentrations such as 2, 4, 6, 8 and 10 μg/mL. The culture suspensions were incubated for 24 h at 37 °C. A loop full of culture was inoculated into MHA medium plates and incubated at 37 °C for 24 h. The MIC and MBC values of AgNPs were documented.

TEM Studies on AgNPs Treated Bacteria

The efficacy of optimized AgNPs on bacterial pathogens was studied using transmission electron microscopy (TEM) (TECNAI 30 G2S-TWIN, FEI Company) [12] and AgNPs were processed as per the method described earlier study [9]. Shortly, 5 μL of bacterial culture (24 h) were treated with AgNPs (10 μg) and incubated shaker with 120 rpm for 24 h at 37 °C. After centrifugation the pellet was collected and examined under TEM.

Antibacterial Activity of Cotton Fabrics Coated AgNPs

The respective MIC of AgNPs was treated with sterilized 1 cm^2 white cotton cloth in a Petri plate. The AgNPs coated cotton was kept in a sterile container at 50 °C for 1 h. The wound infection causing pathogens viz., *B. subtilis, S. aureus, M. luteus, E. coli, K. pneumoniae* and *P. aeruginosa* were swabbed evenly on MHA plates and respective concentrations of AgNPs impregnated cotton

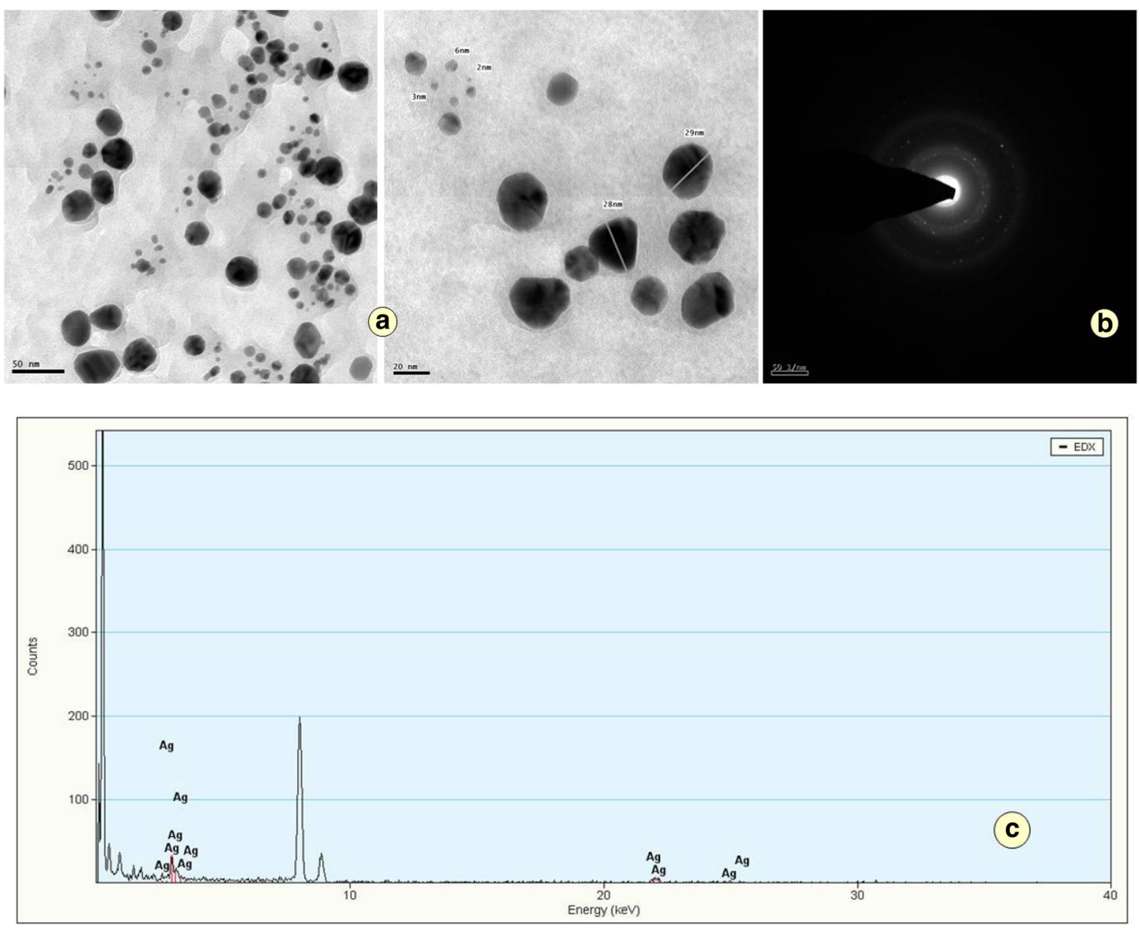

Fig. 1 **a** High resolution transmission electron microscope of optimized AgNPs, **b** selected area electron diffraction of optimized AgNPs, **c** energy dispersive X-ray analysis of optimized AgNPs

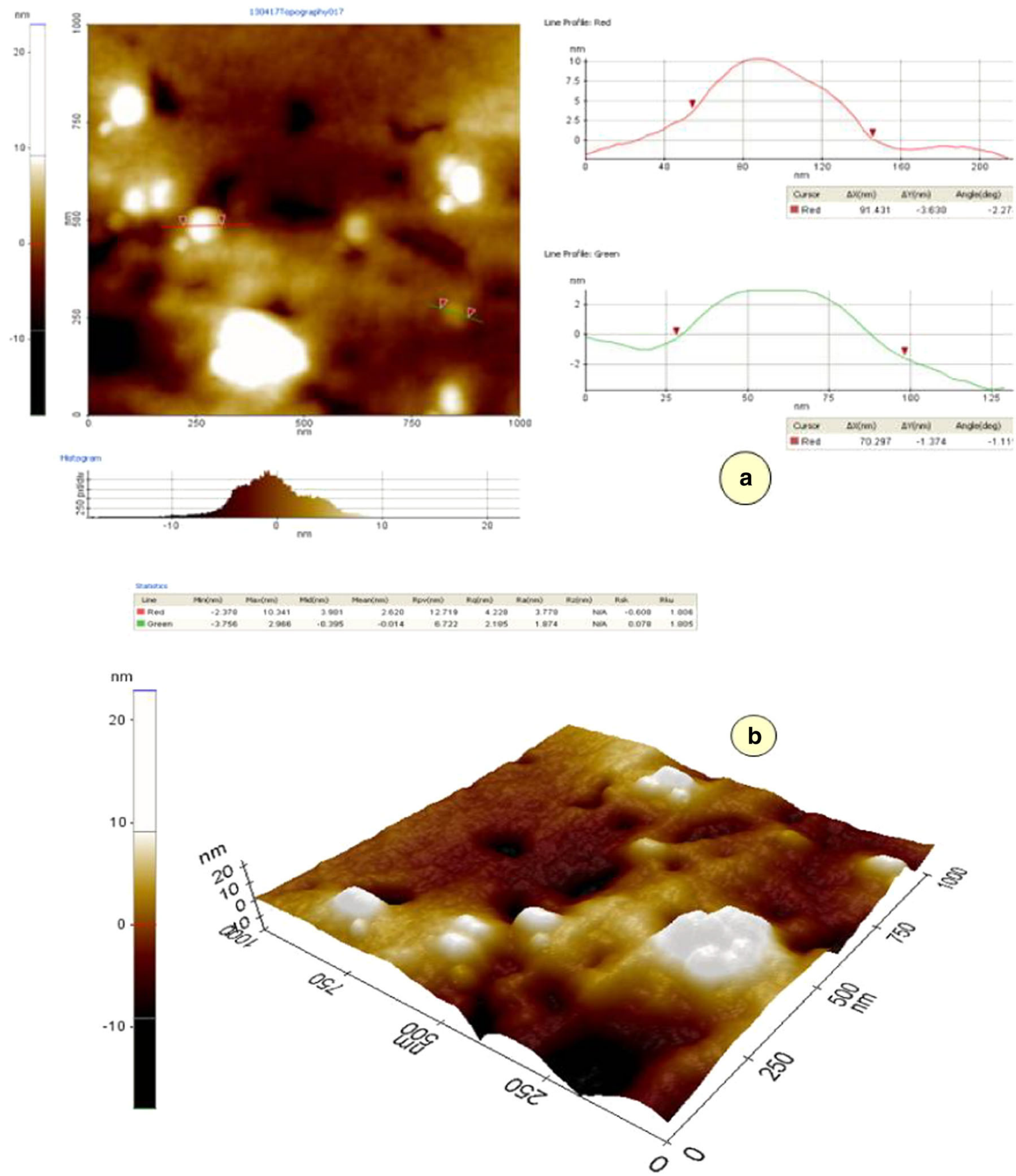

Fig. 2 AFM analysis of optimized phytosynthesized AgNPs

placed center of the plates. The completed plates were incubated at 37 °C for 24 h and zone of inhibition was observed after incubation periods.

Stability of Optimized AgNPs at Different Washings

The stability of MIC AgNPs coated fabrics were recorded for four consecutive washes by using 1 g/L of Surf Excel detergent. The fabric samples were soaked in the detergent water for 10 min and splashed with tap water for 5 min and then dried for 2 h at room temperature. The processes were as accomplished for three subsequent washes, and then they were tested against six bacterial pathogens on the MHA plates. Further, the plates were kept in an incubator for 24 h at 37 °C. Finally, the results were observed and tabulated.

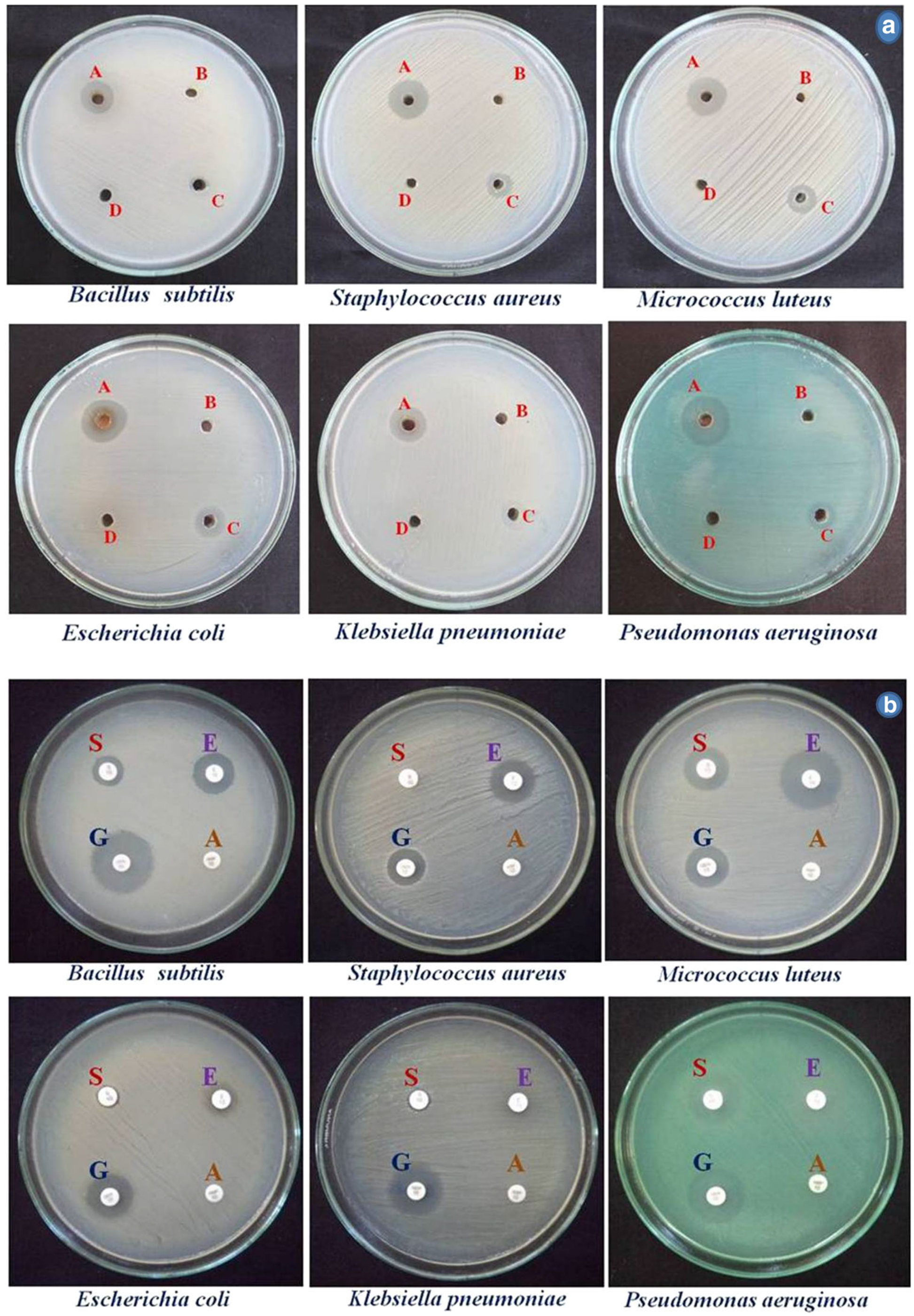

Fig. 3 **a** Antibacterial activity of AgNPs (10 μg) on different human pathogens at 24 h; *A* Phytosynthesized AgNPs, *B* *C. roxburghii* aqueous leaf extract, *C* $AgNO_3$, *D* Sterile GD H_2O. **b** Antibacterial activities of different antibiotics (10 μg) on human pathogens at 24 h; *S* Streptomycin, *G* Gentamycin, *A* Ampicillin, *E* Erythromycin

Thermal Gravimetry Analysis (TGA)

The thermal stability of the optimized and control AgNPs coated cotton fabrics before and after washings was studied using TGA (TA Q50 series) analyzer. This analysis was carried out under N_2 atmosphere at a heating rate of 20 °C min^{-1} at a range of 0–700 °C.

In vitro Cytotoxicity Effect of Optimized AgNPs on Vero Cell Lines—MTT Assay

The cytotoxic activity of optimized AgNPs were tested against common cell line Vero by MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide assay [13]. The Vero cells lines were treated with different concentrations (5, 10, 30, 50, 75 and 100 μg/mL) of silver nitrate and AgNPs. The detailed studies as per the method

Table 1 Effect of optimized AgNPs against different wound infectious human pathogenic bacteria

S. no	Human pathogenic bacteria	Zone of Inhibition (mm)				Positive commercial drug Zone of Inhibition (mm)			
		AgNPs v (10 μg)	$AgNO_3$ (10 μg)	*C. roxburghii* extract (10 μg)	Distilled water (40 μL)	Streptomycin (10 μg)	Gentamycin (10 μg)	Ampicillin (10 μg)	Erythromycin (10 μg)
1	*Bacillus subtilis*	14.6 ± 1.86	7.6 ± 1.06	–	–	11.6 ± 1.15	24.6 ± 1.52	–	17.3 ± 1.52
2	*Staphylococcus aureus*	15.6 ± 1.36	8.3 ± 1.86	–	–	–	15.3 ± 0.57	–	18.6 ± 1.15
3	*Micrococcus luteus*	17.3 ± 1.36	10.0 ± 0.51	–	–	17.0 ± 1.00	15.0 ± 1.00	–	22.0 ± 1.73
4	*Escherichia coli*	17.0 ± 0.89	10 ± 1.36	–	–	–	17.8 ± 0.28	–	8.0 ± 1.00
5	*Klebsiella pneumoniae*	14.0 ± 1.76	8.0 ± 0.89	–	–	–	20.0 ± 1.52	–	8.0 ± 1.00
6	*Pseudomonas aeruginosa*	20.3 ± 1.86	8.6 ± 1.36	–	–	20.6 ± 1.52	23.3 ± 0.57	–	–

"–" No activities

described earlier study [14]. The % of cell sustainability was expressed as:

$$\text{Cell viability}\ (\%) = \frac{\text{Absorbance of treated cells}}{\text{Absorbance of control cells}} \times 100$$

Morphological Changes Analysis

The morphological changes of Vero cell lines were studied, as per method described by our earlier report [15]. Summarily, Vero cells at IC_{50} values were calculated with inverted microscope for untreated and treated cells, after 24 h. The concentration needs to inhibit 50% of cells were referred as IC_{50} value. Finally, the results were observed and photographed.

Results and Discussion

Analysis of HRTEM and SAED of Optimized AgNPs

The optimized AgNPs was spherical in shape with 10–30 nm (Fig. 1a). The optimized AgNPs were confirmed with SAED spectrums that were face-centred cubic of different crystallographic planes (Fig. 1b) [16, 17]. The crystalline natures of the AgNPs were further demonstrated by the SAED pattern with Bragg reflection planes and bright circular spots resultant to (111), (200), (220), (311) and (222) [18]. The EDX spectrum ranges displayed a single solid signal for silver (Fig. 1c), it designated that the synthesized AgNPs were absolute and free of and our results synchronizes with the earlier observations made on [19].

Atomic Force Microscopy (AFM)

External morphology of the AgNPs was studied under AFM as shown in Fig. 2a. The sizes of the particles are in the range of ~ 70–90 nm for AgNPs, the respective 3D images (AFM) are displayed in Fig. 2b. The AgNPs were spherical in shape with a smooth surface, without any pits and clefts. The spherical shape of AgNPs was noticed in the existent efforts [20]. The size of the AgNPs was 28 nm (*Ocimum tenuiflorum*), 26.5 nm (*Syzygium cumini*), 65 nm (*Camellia sinensis*), 22.3 nm (*Solanum tricobatum*) and 28.4 nm (*Colubrina asiatica*) [21].

Antibacterial Activity of Optimized AgNPs Against Wound Infection Causing Pathogenic Bacteria

The effect of the antibacterial property of optimized phytosynthesized AgNPs was made versus three different gram positive and gram negative wound infection causing pathogens. Among, the evaluated pathogens, *P. aeruginosa* were found to be highly sensitive to the optimized AgNPs showing 20.3 ± 1.86 mm of zone of inhibition followed by *M. luteus* of 17.3 ± 1.36 mm and *E. coli* of 17.0 ± 0.89 mm, *S. aureus* of 15.6 ± 1.36 mm, *B. subtilis* of 14.6 ± 1.86 mm and *K. pneumoniae* of 14.0 ± 1.76 mm. The aqueous extract of *C. roxburghii* and sterile distilled water not shown any inhibitory activity alongside bacterial pathogens and the silver nitrate ($AgNO_3$) alone exhibited least activity (Fig. 3a, Table 1). The antibiotic, gentamycin showed the activity of

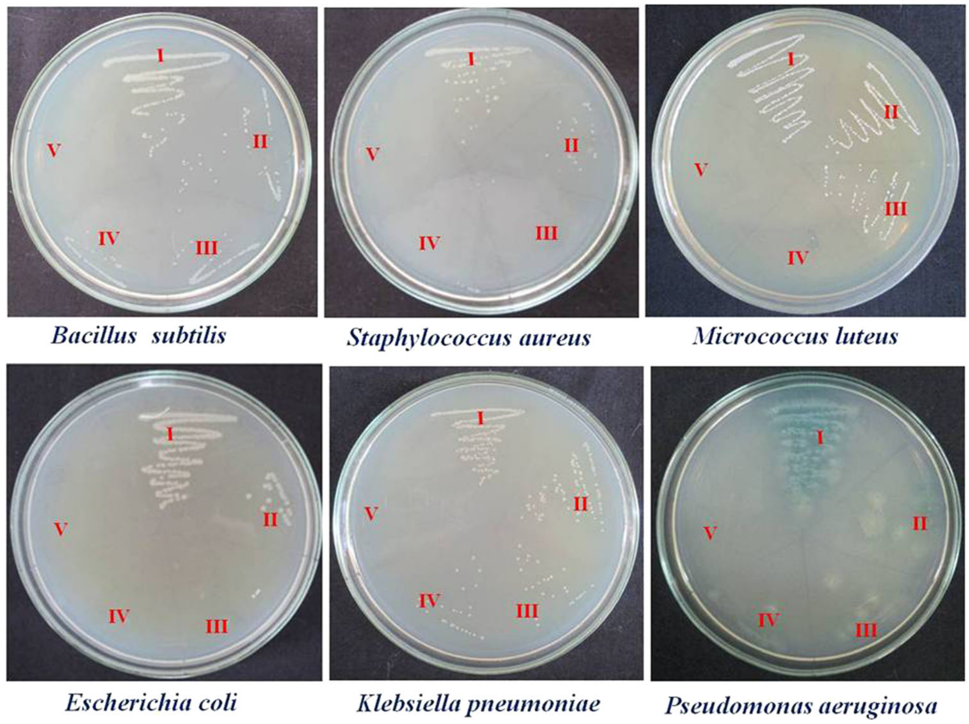

Fig. 4 Minimum inhibitory concentration and minimum bactericidal concentration of optimized AgNPs against wound infectious pathogens at 24 h

Table 2 MIC and MBC of optimized AgNPs against different human pathogenic bacteria

S. no	Human pathogenic bacteria	MIC (µg/mL)	MBC (µg/mL)
1	*Bacillus subtilis*	8.1 ± 0.76	9.8 ± 0.23
2	*Staphylococcus aureus*	4.0 ± 1.00	6.3 ± 0.47
3	*Micrococcus luteus*	6.3 ± 0.57	7.8 ± 0.62
4	*Escherichia coli*	4.0 ± 1.00	6.3 ± 0.47
5	*Klebsiella pneumoniae*	8.1 ± 0.76	9.8 ± 0.23
6	*Pseudomonas aeruginosa*	6.3 ± 0.57	8.1 ± 0.23

24.6 ± 1.52 mm zone of inhibition towards *B. subtilis* continued by *P. aeruginosa* of 23.3 ± 0.57 mm, *K. pneumoniae* of 20.0 ± 1.52 mm *E. coli* of 17.8 ± 0.28 mm, *S. aureus* of 15.3 ± 0.57 mm and *M. luteus* of 15.0 ± 1.00 mm. Whereas the antibiotic, erythromycin exhibited high activity against *M. luteus* of 22.0 ± 1.73 mm and less activity towards *K. pneumoniae* and *E. coli* of each 8.0 ± 1.00 mm, and nil activity towards *P. aeruginosa*. The other antibiotic, streptomycin showed good activity against *P. aeruginosa* of 20.6 ± 1.52 mm and less activity towards *B. subtilis* of 11.6 ± 1.15 mm and it did not show efficacy towards *S. aureus, E. coli* and *K. pneumoniae*. The tested pathogens were resistant to the antibiotic ampicillin (Fig. 3b, Table 1). The precise antimicrobial activity of AgNPs is yet to clear, however reports suggested possible three different mechanisms of action, which involves the adherence of silver ions causing plasmolysis to the cell membrane [22] and the silver nanoparticles are strong interaction with DNA particularly in the sulfur–phosphorus molecules, outer membrane proteins of bacterial cells thus, disturbing the oxidative phosphorylation reactions [23]. Convincingly, AgNPs also release silver ions and it penetrates into the cell wall and impairment the DNA by condensation and it also affecting the protein synthesis [24].

MIC and MBC of Optimized AgNPs Against Bacterial Pathogens

The MIC of AgNPs was documented versus six varying bacteria. The MIC value of *B. subtilis* and *K. pneumoniae* was at each 8.1 ± 0.76 µg/mL and *M. luteus* and *P. aeruginosa* was at each 6.3 ± 0.57 µg/mL and *S. aureus* and *E. coli* were at each 4.0 ± 1.00 µg/mL [25]. Among the experimental bacteria, *S. aureus* and *E. coli* were found highly sensitive to the phytosynthesized AgNPs and *M. luteus* and *P. aeruginosa* showed moderate resistance than *B. subtilis* and *K. pneumoniae* (Fig. 4). Whereas the MBC of phytosynthesized AgNPs were 6 µg/mL each towards *S. aureus* and *E. coli* 7.8 ± 0.62 µg/mL for *M. luteus* and 8.1 ± 0.23 for *P. aeruginosa* and 9.8 ± 0.23 µg/mL each for *B. subtilis* and *K. pneumoniae* (Fig. 4, Table 2). The AgNPs treated bacteria such as *E. coli, P. aeruginosa, S. aureus, Klebsiella* sp. and *B. subtilis* showed 6–12 µg/mL, 12–50 µg/mL, 25–50 µg/mL, 6–25 µg/mL and 25–50 µg/mL for MIC–MBC, respectively [26].

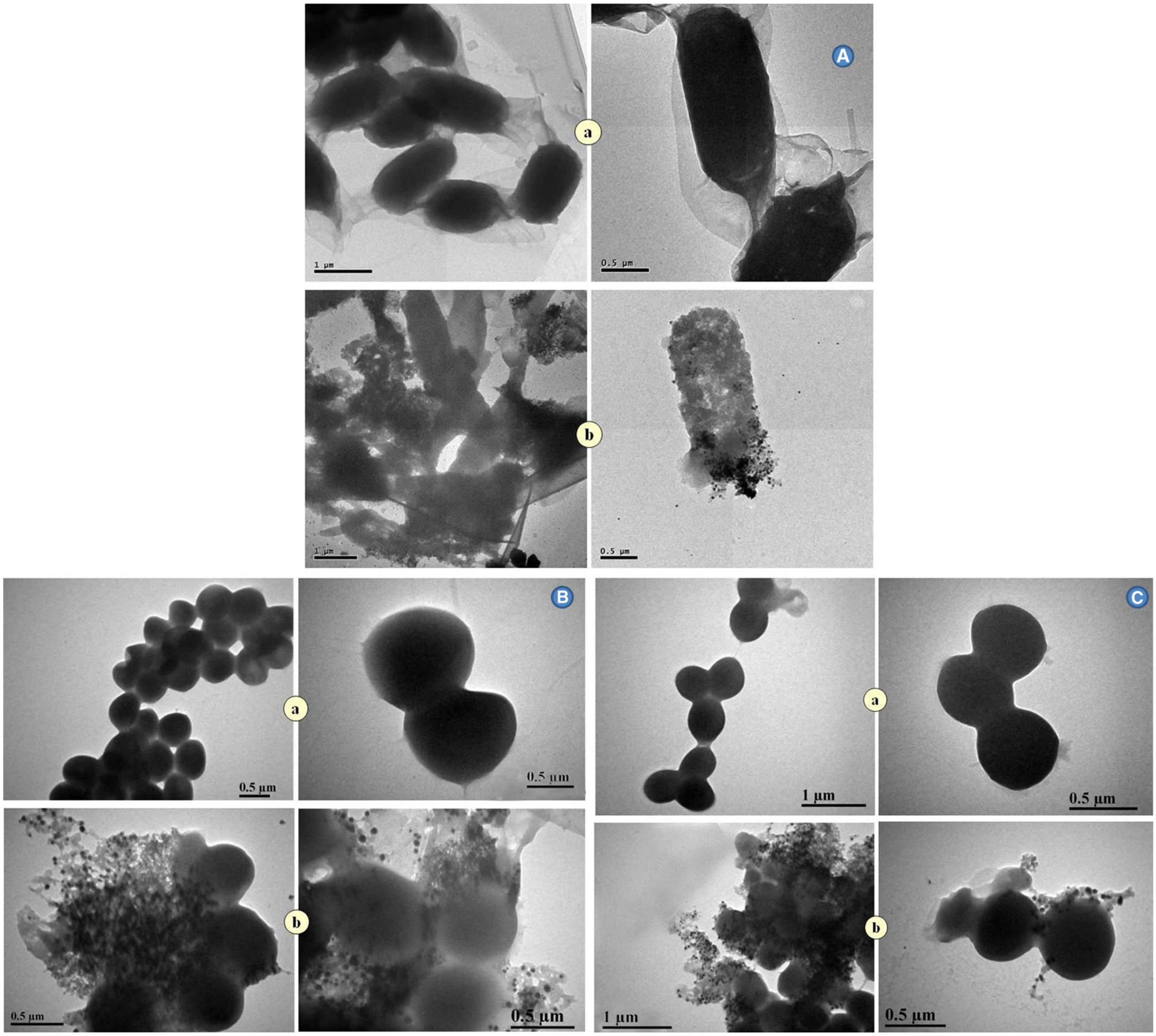

Fig. 5 **a** TEM images of (*A*) *B. subtilis*, (*B*) *S. aureus*, (*C*) *M. luteus*. interactions with AgNPs, *a* Control, *b* AgNPs treated bacterial cells. **b** TEM images of (*D*) *E. coli*, (*E*) *K. pneumoniae*, (*F*) *P. aeruginosa* interactions with AgNPs; *a* Control, *b* AgNPs treated bacterial cells

TEM Studies on Optimized AgNPs Treated Bacteria

The optimized AgNPs treated *B. subtilis, S. aureus* and *M. luteus* cells after 24 h was analyzed using TEM, which showed adherence of the AgNPs on the periphery and also embedded in the cells resulting in the bacterial cell lysis and releasing their cell contents in the surrounding area (Fig. 5a, b, c). Similarly, *B. subtilis, S. aureus* and *M. luteus* treated cells also showed cell lysis by exhibiting broken cell walls and membranes. The TEM analysis of three gram-negative bacteria: *E. coli, K. pneumoniae* and *P. aeruginosa* cells treated with optimized AgNPs at 24 h showed that the cells are found aberrant, cracked, ruptured and forming pits in the cell membrane (Fig. 5d, e, f). Leakages of cell contents of AgNPs treated cells were observed through bacterial cell wall denaturation, interfering of bacterial respiration, cell membrane weakening and ATP exhaustion [27]. Sondi and Salopek-Sondi [6] also studied that AgNPs were aggregated in the cell membrane defeating their permeability thus causing cell death. Interestingly, despite this permeability obstruction, in the current study, the AgNPs displayed a substantial inhibitory activity against both bacterial pathogens.

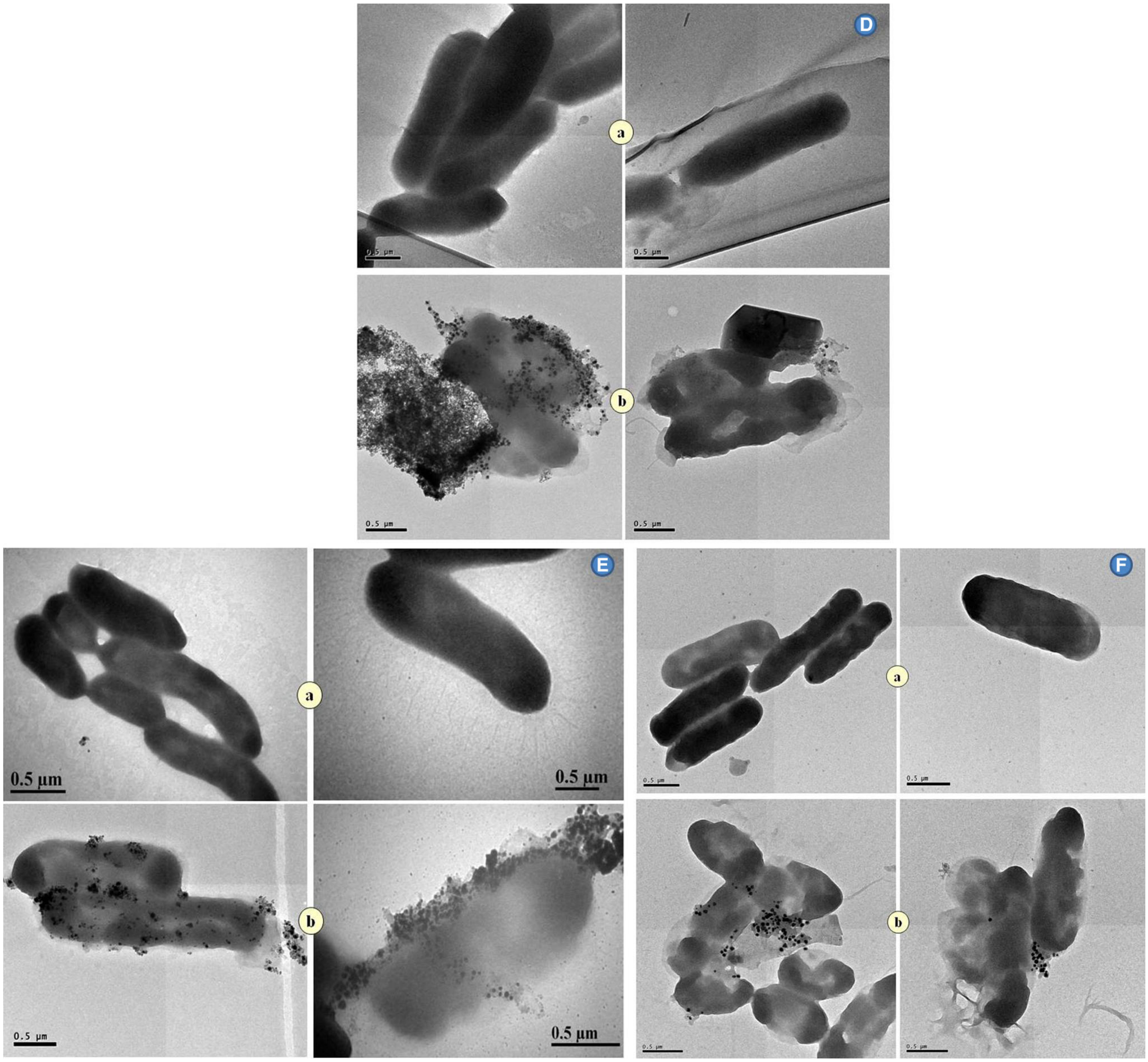

Fig. 5 continued

Antibacterial Activity of Optimized AgNPs Coated Cotton Fabrics

The cotton fabric samples impregnated with optimized AgNPs of respective minimum inhibitory concentration against the six different bacteria pathogens were investigated. Among, *P. aeruginosa* was found extremely sensitive to AgNPs by showing 18.1 ± 0.76 mm of zone of inhibition followed by *S. aureus* of 18.1 ± 0.25 mm, *E. coli* of 16.8 ± 1.04 mm, *M. luteus* of 16.3 ± 0.57 mm, *B. subtilis* of 14.3 ± 0.57 mm, and *K. pneumonia* of 12.3 ± 0.57 mm. The non-AgNPs coated cotton fabrics did not show activity against the tested bacterial pathogens (Fig. 6a, Table 3). The AgNPs loaded fabrics showed good antibacterial activity but fabric without AgNPs showed null activity [28]. The cloth immobilized with AgNPs possesses greater antibacterial activity against *E. coli* whereas no activity was recorded without AgNPs immobilized cloth [29].

The cotton cloth restrained by AgNPs could be applied as a disinfectants dressing material for wound plaster against pathogens [30]. According to Rifaya and Meyyappan [31] reports "these bacteria have been shown to have longer survival times on regularly used hospital draperies, like hospital privacy drapes, scrub suits, and lab coats". Cotton fibres are very popular and people extensively useful in routine life because of their exceptional properties *i.e.*, softness, hygroscopicity, skin compatibility,

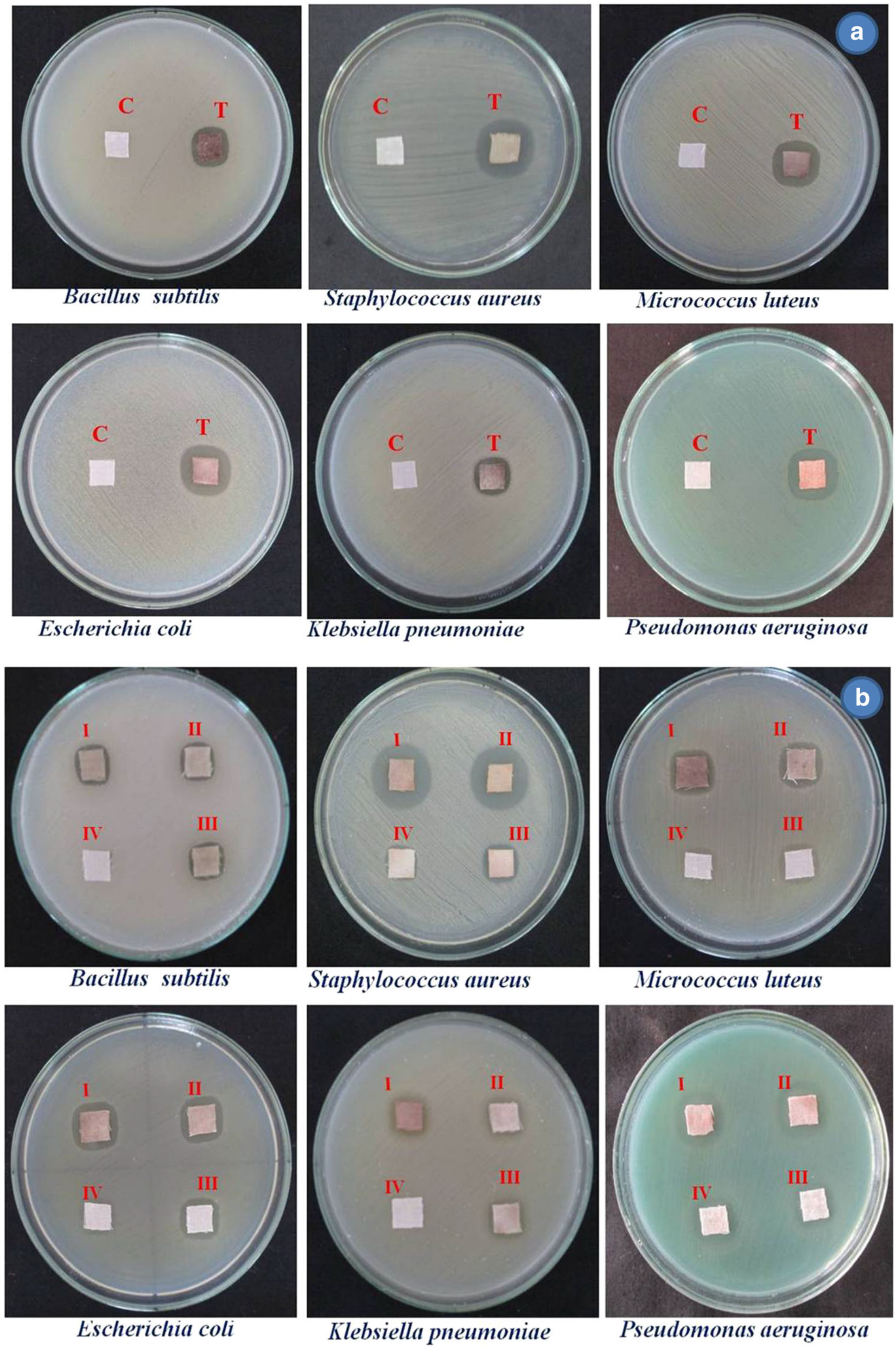

Fig. 6 **a** Effect of MIC—AgNPs impregnated on cotton fabrics; C—Control (Cotton fabric without AgNPs), T—Test (MIC- Cotton fabric impregnated with AgNPs). **b** Washings of AgNPs impregnated cotton fabric against different wound infectious human pathogens at 24 h; 1st Washing-I, 2nd Washing-II, 3rd Washing-III, 4th Washing-IV

sweat absorbing properties and regeneration property [32]. The AgNPs saturated antiseptic cloth may be a good suitability in briefs and socks [30].

Stability of Optimized AgNPs at Different Washings

The chances of endurance and transmission of microbes among the patients and the healthcare workers have been recorded [33] and thus shows the demand for such antimicrobial fabrics [3]. The collective customers'

Table 3 Effect of four consecutive washes on AgNPs—MIC impregnated cotton fabrics on different bacteria at 24 h

S. no.	Human pathogenic bacteria	Zone of Inhibition (mm)				
		Before wash	Wash-1	Wash-2	Wash-3	Wash-4
1	*Bacillus subtilis*	14.3 ± 0.57	14.3 ± 1.52	14.1 ± 0.76	12.1 ± 1.25	9.83 ± 0.28
2	*Staphylococcus aureus*	18.1 ± 0.25	18.1 ± 0.16	18.0 ± 1.00	15.0 ± 1.00	12.0 ± 1.00
3	*Micrococcus luteus*	16.3 ± 0.57	16.1 ± 1.25	15.8 ± 0.76	13.1 ± 1.25	11.8 ± 0.28
4	*Escherichia coli*	16.8 ± 1.04	16.1 ± 0.76	15.66 ± 0.76	13.88 ± 0.28	11.6 ± 0.57
5	*Klebsiella pneumoniae*	12.3 ± 0.57	12.1 ± 1.25	11.1 ± 1.25	9.83 ± 0.76	–
6	*Pseudomonas aeruginosa*	18.1 ± 0.76	17.3 ± 0.28	15.1 ± 1.25	12.3 ± 0.57	9.8 ± 0.76

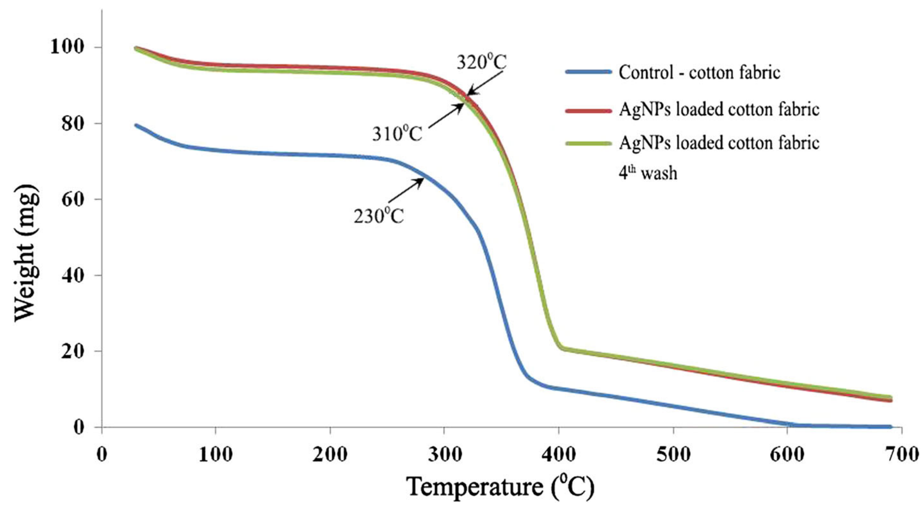

Fig. 7 Thermal gravimetric analysis (TGA) of cotton fabric impregnated with optimized AgNPs

responsiveness with regards to health, hygiene, makes the demand for antibacterial textiles. Consequently, various antibacterial agents have been used to fabricate antimicrobial knits [34].

In the present attempt optimized AgNPs impregnated on the cotton fabrics were washed with laundry detergent for four consecutive washes. The antibacterial activity of the sample after the first wash was found to be maximum of 18.1 ± 0.16 mm zone of inhibition against *S. aureus* followed by *P. aeruginosa* with 17.3 ± 0.28 mm. There was a gradual decrement in the antibacterial activity against the test organism at 2nd and 3rd wash. The efficacy of the AgNPs was reduced up to 40% at the end of 4th wash against *P. aeroginosa* and it was nil against *K. pneumoniae*. Whereas the efficacy was reduced up to 28%, 33%, 25% and 25% against *B. subtilis*, *S. aureus*, *M. luetus* and *E. coli,* respectively, when compared to the first wash (Fig. 6b, Table 3). The present attempt was following with the earlier study [35]. Washing of the fabric coated with AgNPs showed a decrement in the antibacterial activity and the prime reason for testing the fabrics by washing being that these fabrics can be repeatedly used in the hospital [36].

Thermal Gravimetry Analysis (TGA)

Figure 7 illustrates the thermal degradation of (i) cotton fabric without AgNPs (ii) cotton fabric with optimized AgNPs and (iii) optimized AgNPs loaded cotton fabric after 4th wash. Initially, there was a weight loss up to 100 °C. The samples showed a two-stage weight loss firstly, the cotton fabric sample of (i) exhibited at 230 °C and the samples of (ii) and (iii) at 320 °C and 310 °C, respectively. Secondly, the cotton fabric samples of (i) at 380 °C and the samples (ii) and (iii) at each 400 °C. Interestingly even after 4th wash the optimized phytosynthesized AgNPs loaded samples showed similar values to that of unwashed samples (ii) of cotton fabric with AgNPs. Silver nanoparticles coated cotton fabrics after 4th wash showed increased stability than that of the fabric without AgNPs. This could be due to the adherence of AgNPs in the fabric. Hosseini and Ali Pairovi [37] observed that washing increased the stability of the fabric as the unreacted acid and oligomers adhering to the surface of fabric are washed away or neutralized by this treatment stimulating better lifetime and stability of the AgNPs coated fabric [38]. Degradation of the washed fabrics started at a

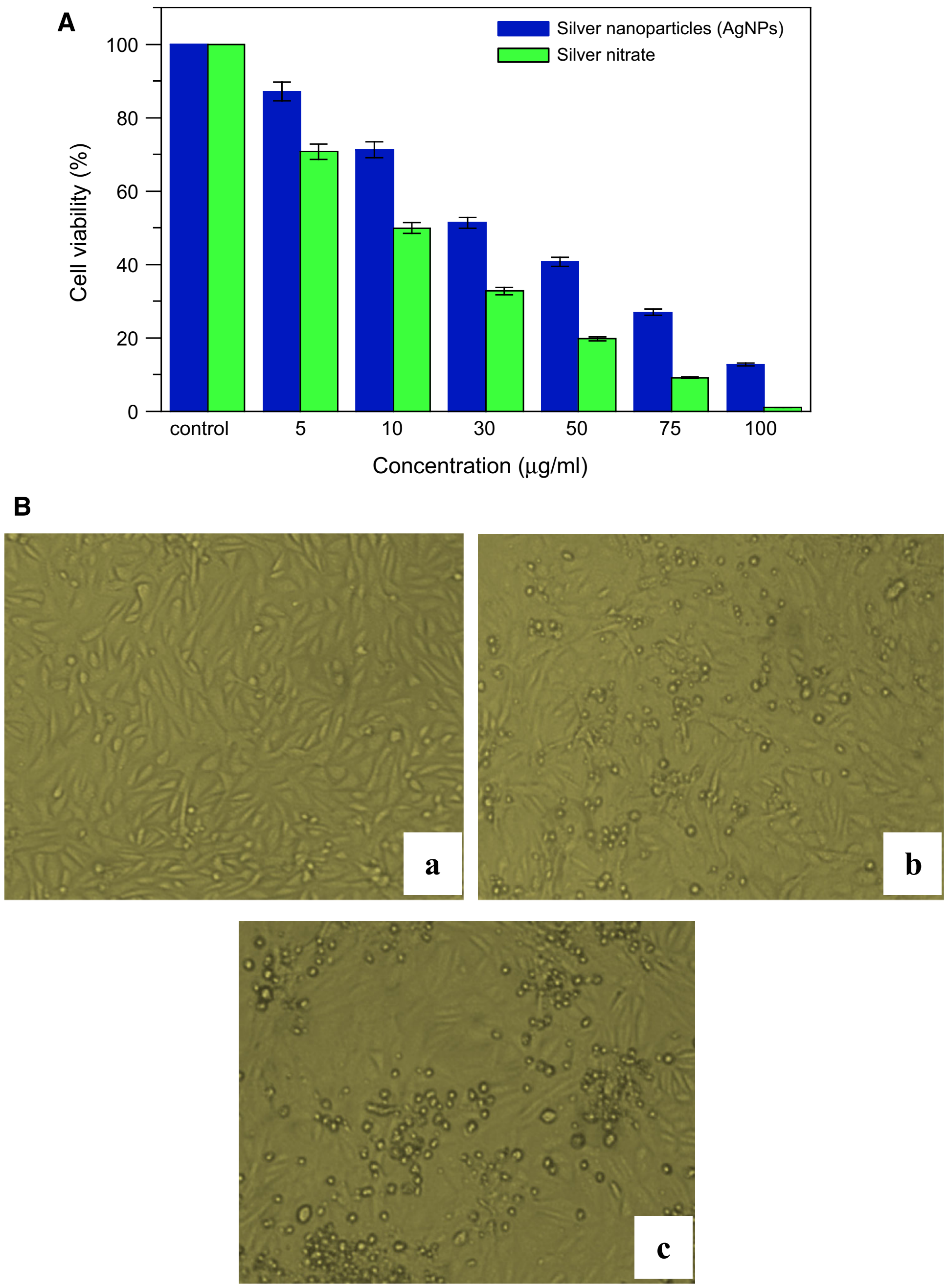

Fig. 8 **A** Effect of $AgNO_3$ and optimized AgNPs on Vero cell line at 24 h; **a** Control, **b** optimized AgNPs treated (20 µg/mL), **c** $AgNO_3$ treated (5 µg/mL). **B** Effect of different concentration of $AgNO_3$ and optimized AgNPs on Vero cell line at 24 h. Values are given as mean ± S.D. for each concentration. The treated concentrations statistically significant ($p < 0.05$) when compared to control

higher temperature than the other samples since washing caused the fibre filament to swell. This is due to the partial separation of the fibre by the molecule of water [38].

Assessment of In Vitro Cytotoxicity of Vero (Normal) Cell Lines

The in vitro cytotoxicity effects of optimized AgNPs treated cells showed IC_{50} value 20 µg/mL, whereas $AgNO_3$ 5 µg/mL (Fig. 8A). Optimized AgNPs tested Vero cells were recorded low toxicity than the $AgNO_3$. The cell viabilities suggestively were reduced in AgNPs treated cell when compared to untreated ($p < 0.05$). The $AgNO_3$ and AgNPs treated Vero cells with IC_{50} concentration were found to be shirked and rounded (Fig. 8B). About cell death, a minimum of AgNPs at 30 µg/mL was well enough to induce 50% cell mortality [39]. The cytotoxicity study confirmed that, AgNPs are not chronically toxic to the cell growth and their viability. The passage of AgNPs inside the cell begins with recognition of membrane receptor, internalization, translocation, and ends with degradation, accumulation, or clearance by cells [40].

Conclusion

The comprehensive results suggested that the antibacterial activity of AgNPs showed much potent inhibitory activity, as shown by their MIC value and can be used as a silver plaster of wounds or as AgNPs impregnated cotton fabrics. The cotton fabrics, having excellent antibacterial properties and can withstand repeated washing. The antibacterial activity of the normal morphology of gram positive and gram negative bacteria was changed when treated with AgNPs. The cells showed aberrant morphology and they were cracked and ruptured as evidenced under TEM. In vitro cytotoxicity assay also confirmed the optimized AgNPs are less toxic. We conclude that the effective antibacterial activity of biologically synthesized AgNPs reported in this study can be exploited as an antibacterial agent with submissions in therapeutic treatments. The inclusion of these AgNPs can avoid the proliferation of opportunistic pathogens associated with infections in and out patients. Hence, the optimized AgNPs might be used as commendable therapeutic material against microbial pathogens and also for the successful development of drug delivery in near future. The data represented in our study contribute to a novel and unexplored area of nano-materials for alternative medicine. In addition, future study is going to investigation of toxicity effect with different cell lines and also wound repair activities against mesenchymal stem cell and/or fibroblast (NIH3T3) cells regarding their short-term and long-term toxicity.

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Compliance with Ethical Standards

Conflicts of interest The authors report no conflicts of interest in performing this work.

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